Modality

The process of **modality** is where a device, receiving information from a user through sensors and inputs, is expected to perform multiple actions based on that input. In our last example, the interaction was fairly simple, with a button being pressed five times to trigger the activation of the LED. What if we wanted multiple actions to occur based on multiple types of input? To do this we could use the `switch` statement and a counter based on the last example. Listing 5-4 is one possibility.

Listing 5-4. *State Change Example 3*

```
const int switchPin = 2;
const int ledPin = 13;

int state = 0;
int lastState = 0;
int buttonCounter = 0;

unsigned long startTime = 0;
unsigned long interval = 500;

void setup() {
  pinMode(ledPin, OUTPUT);
  pinMode(switchPin, INPUT);
}

void loop() {
  state = digitalRead(switchPin);
  if(state != lastState) {
    if (state == HIGH) {
      buttonCounter++;
      startTime = millis();
    }
  }
  lastState = state;

  if(startTime + interval < millis()) {

    switch (buttonCounter) {
    case 1:
      digitalWrite(ledPin, HIGH);
      delay(interval);
      digitalWrite(ledPin, LOW);
      delay(interval);
      break;
```